

B200				GH200			MI300X
	Triton	cuTile	TileLang	Triton	cuTile	TileLang	Triton
matmul · fp32	TMA TMEM+SMEM	TMA TMEM+SMEM	cp.async SMEM	TMA SMEM	TMA SMEM	cp.async SMEM	desc → ptr LDS
matmul · fp16	TMA TMEM+SMEM	TMA TMEM+SMEM	cp.async TMEM+SMEM	TMA SMEM	TMA SMEM	cp.async SMEM	desc → ptr LDS
matmul · fp8_e4m3fn	TMA TMEM+SMEM	TMA TMEM+SMEM	cp.async TMEM+SMEM	TMA SMEM	TMA SMEM	cp.async SMEM	unsupported
matmul_int8 · int8	TMA SMEM	TMA TMEM+SMEM	cp.async+LDG TMEM+SMEM	TMA SMEM	TMA SMEM	cp.async+LDG SMEM	desc → ptr LDS
batched_matmul · fp32	TMA TMEM+SMEM	TMA TMEM+SMEM	cp.async SMEM	TMA SMEM	TMA SMEM	cp.async SMEM	desc → ptr LDS
streamk_matmul · fp16	TMA TMEM+SMEM	TMA TMEM+SMEM	cp.async TMEM+SMEM	TMA SMEM	TMA SMEM	cp.async SMEM	desc → ptr LDS
flash_attention · fp16	TMA TMEM+SMEM	TMA TMEM+SMEM	LDG TMEM+SMEM	TMA SMEM	TMA SMEM	LDG SMEM	desc → ptr LDS
block_sparse_attn · fp16	TMA+LDG SMEM	TMA+LDG TMEM+SMEM	LDG TMEM+SMEM	TMA+LDG SMEM	TMA+LDG SMEM	LDG SMEM	desc → ptr LDS
linear_self_attn · fp32	TMA+LDG TMEM+SMEM	TMA+LDG TMEM+SMEM	LDG SMEM	TMA+LDG SMEM	TMA+LDG SMEM	LDG SMEM	desc → ptr LDS
1d_conv · fp16	LDG TMEM+SMEM	LDG SMEM	LDG TMEM+SMEM	LDG SMEM	LDG SMEM	LDG SMEM	pointer LDS
3d_conv · fp32	cp.async TMEM+SMEM	LDG TMEM+SMEM	LDG SMEM	cp.async SMEM	cp.async SMEM	LDG SMEM	pointer LDS
destindex · int8	LDG –	LDG SMEM	LDG –	LDG –	LDG SMEM	LDG –	pointer –
histogramming · int32	LDG SMEM	TMA+LDG SMEM	LDG SMEM	LDG SMEM	TMA SMEM	LDG SMEM	pointer LDS
2d_max_pooling · fp32	LDG SMEM	LDG SMEM	LDG –	LDG SMEM	LDG SMEM	LDG –	pointer –
vector_add · fp32	LDG –	LDG –	LDG –	LDG –	LDG –	LDG –	pointer –

Top line: operand load path (TMA, cp.async, LDG, desc → ptr = TensorDescriptor lowered to pointer loads, pointer).

Bottom line: on-chip staging (SMEM, TMEM, LDS; – = none observed).

tcgen05
 legacy MMA (HMMA/IMMA)
 no matrix instruction
 unsupported

WGMMMA
 MFMA

static SASS/ISA only